

Vanilla GCN: Mathematical Modelling

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Detecting illicit Bitcoin transactions on the Elliptic dataset using a 2-layer Vanilla GCN.

1. Graph Representation

$$G = (V, E), \quad |V| = 203,769 \text{ (transactions)}, \quad |E| = 234,355 \text{ (money flows)}$$

Adjacency matrix $A \in \{0,1\}^{N \times N}$, feature matrix $X \in \mathbb{R}^{N \times F}$ ($F = 165$), label $y \in \{0,1\}$ (0 = licit, 1 = illicit).

2. Symmetric Normalisation

$$\tilde{A} = A + I_N, \quad \tilde{D}_{ii} = \sum_j \tilde{A}_{ij}, \quad \hat{A} = \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}$$

Self-loops let a node keep its own features. Normalisation gives a fair weighted average, so high-degree nodes do not dominate.

Feature normalisation: $x_{ij} \leftarrow \frac{x_{ij} - \mu_j}{\sigma_j}$ (mean μ_j , standard deviation σ_j of feature j).

3. GCN Layer and Forward Pass

$$H^{(l+1)} = \sigma(\hat{A} H^{(l)} W^{(l)})$$

$H^{(l)}$ = node features at layer l , $W^{(l)}$ = learnable weights, $\hat{A}$ = normalised adjacency, σ = ReLU.

$$Z = \text{softmax}(\hat{A} \cdot \text{ReLU}(\hat{A} \cdot X \cdot W^{(0)}) \cdot W^{(1)})$$

$N = 203,769$, $F = 165$, $H = 100$ hidden, $C = 2$ classes: $X \rightarrow N \times F$, $H^{(1)} \rightarrow N \times H$, $Z \rightarrow N \times C$

4. Loss Function

Weighted cross-entropy over labelled training nodes only (illicit is rare, so $w_1 = 0.7$, $w_0 = 0.3$):

$$\mathcal{L} = -\frac{1}{\sum_i w_{y_i}} \sum_{i \in V_{\text{train}}} w_{y_i} \log Z_{i,y_i}$$

5. Training

Minimise $\mathcal{L}$ over $\theta = \{W^{(0)}, W^{(1)}\}$ with Adam: $\theta \leftarrow \theta - \eta \cdot \text{Adam}(\nabla_{\theta} \mathcal{L})$, $\eta = 0.01$, 200 epochs.

One epoch: forward pass $\rightarrow$ loss on train nodes $\rightarrow$ backpropagation $\rightarrow$ update weights. Train on time steps 1–34, test on 35–49.

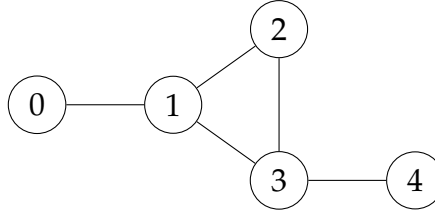
6. Evaluation Metrics

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad F_1 = \frac{2 \cdot P \cdot R}{P + R}$$

Our result ($TP = 507, FP = 182, FN = 576$): $\text{Precision} = \frac{507}{689} = 0.736$, $\text{Recall} = \frac{507}{1083} = 0.468$, $F_1 = 0.572$.

7. Worked Example: Graph

Five transactions, $V = \{0, 1, 2, 3, 4\}$, $E = \{(0, 1), (1, 2), (1, 3), (2, 3), (3, 4)\}$.



$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}, \quad d = [1, 3, 2, 3, 1], \quad D = \text{diag}(1, 3, 2, 3, 1)$$

Add self-loops:

$$\tilde{A} = A + I = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}, \quad \tilde{d} = [2, 4, 3, 4, 2], \quad \tilde{D} = \text{diag}(2, 4, 3, 4, 2)$$

8. Normalised Adjacency

$$\tilde{D}^{-1/2} = \text{diag}\left(\frac{1}{\sqrt{2}}, \frac{1}{2}, \frac{1}{\sqrt{3}}, \frac{1}{2}, \frac{1}{\sqrt{2}}\right), \quad \hat{A} = \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}$$

$$\hat{A}_{ij} = \frac{\tilde{A}_{ij}}{\sqrt{\tilde{d}_i \tilde{d}_j}} \Rightarrow \hat{A} \approx \begin{bmatrix} 0.5 & 0.354 & 0 & 0 & 0 \\ 0.354 & 0.25 & 0.289 & 0.25 & 0 \\ 0 & 0.289 & 0.333 & 0.289 & 0 \\ 0 & 0.25 & 0.289 & 0.25 & 0.354 \\ 0 & 0 & 0 & 0.354 & 0.5 \end{bmatrix}$$

e.g. $\hat{A}_{01} = \frac{1}{\sqrt{2 \cdot 4}} = \frac{1}{2\sqrt{2}} \approx 0.354$, $\hat{A}_{12} = \frac{1}{\sqrt{4 \cdot 3}} = \frac{1}{2\sqrt{3}} \approx 0.289$.

9. Features and Labels

f_1 = unusual output pattern, f_2 = normal fee pattern.

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \quad \begin{array}{l} \text{Node 0: illicit, } y_0 = [0, 1] \\ \text{Node 4: licit, } y_4 = [1, 0] \\ \text{Nodes 1, 2, 3: unlabelled} \end{array}$$

10. Layer 1: $H^{(1)} = \text{ReLU}(\hat{A}XW^{(0)})$

Take $W^{(0)} = I$ (only neighbour mixing), so $H^{(1)} = \text{ReLU}(\hat{A}X)$:

$$\begin{aligned} H_0^{(1)} &= 0.5(1, 0) + 0.354(1, 1) = (0.854, 0.354) \\ H_1^{(1)} &= 0.354(1, 0) + 0.25(1, 1) + 0.289(1, 0) + 0.25(0, 1) = (0.892, 0.5) \\ H_2^{(1)} &= 0.289(1, 1) + 0.333(1, 0) + 0.289(0, 1) = (0.622, 0.577) \\ H_3^{(1)} &= 0.25(1, 1) + 0.289(1, 0) + 0.25(0, 1) + 0.354(0, 1) = (0.539, 0.854) \\ H_4^{(1)} &= 0.354(0, 1) + 0.5(0, 1) = (0, 0.854) \end{aligned}$$

All values are ≥ 0 , so ReLU makes no changes:

$$H^{(1)} \approx \begin{bmatrix} 0.854 & 0.354 \\ 0.892 & 0.500 \\ 0.622 & 0.577 \\ 0.539 & 0.854 \\ 0 & 0.854 \end{bmatrix}$$

11. Layer 2: $Z = \hat{A}H^{(1)}W^{(1)}$

$$W^{(1)} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}, \quad M = \hat{A}H^{(1)}, \quad Z = MW^{(1)}$$

$$\begin{aligned} M_0 &= 0.5(0.854, 0.354) + 0.354(0.892, 0.5) = (0.742, 0.354) \Rightarrow Z_0 = (-0.389, 0.389) \\ M_4 &= 0.354(0.539, 0.854) + 0.5(0, 0.854) = (0.190, 0.729) \Rightarrow Z_4 = (0.538, -0.538) \end{aligned}$$

Same for all rows ($Z_{i0} = M_{i1} - M_{i0}$, $Z_{i1} = -Z_{i0}$):

$$M \approx \begin{bmatrix} 0.742 & 0.354 \\ 0.839 & 0.630 \\ 0.620 & 0.583 \\ 0.537 & 0.807 \\ 0.190 & 0.729 \end{bmatrix} \quad Z \approx \begin{bmatrix} -0.389 & 0.389 \\ -0.209 & 0.209 \\ -0.037 & 0.037 \\ 0.270 & -0.270 \\ 0.538 & -0.538 \end{bmatrix}$$

12. Softmax

$$P_{ic} = \frac{e^{Z_{ic}}}{\sum_{c'} e^{Z_{ic'}}}$$

$$\text{Node 0: } P_{0,1} = \frac{e^{0.389}}{e^{-0.389} + e^{0.389}} \approx 0.685 \Rightarrow P_0 = [0.315, 0.685] \rightarrow \text{illicit } \checkmark$$

$$\text{Node 4: } P_{4,0} = \frac{e^{0.538}}{e^{0.538} + e^{-0.538}} \approx 0.746 \Rightarrow P_4 = [0.746, 0.254] \rightarrow \text{licit } \checkmark$$

$$P \approx \begin{bmatrix} 0.315 & 0.685 \\ 0.397 & 0.603 \\ 0.481 & 0.519 \\ 0.632 & 0.368 \\ 0.746 & 0.254 \end{bmatrix} \quad \begin{array}{l} \text{Node 0: illicit } \checkmark \\ \text{Node 1: illicit (predicted)} \\ \text{Node 2: illicit (borderline)} \\ \text{Node 3: licit (predicted)} \\ \text{Node 4: licit } \checkmark \end{array}$$

Unlabelled node 1 is flagged illicit because it receives money directly from the illicit node 0.

13. Cross-Entropy Loss

$$\mathcal{L} = \frac{1}{2} [-\log P_{0,1} - \log P_{4,0}] = \frac{1}{2} [-\log 0.685 - \log 0.746] = \frac{0.378 + 0.293}{2} \approx \underline{\underline{0.336}}$$

14. Gradient and Update

$$\frac{\partial \mathcal{L}_i}{\partial Z_{ic}} = P_{ic} - y_{ic}$$

$$\text{Node 0: } [0.315 - 0, 0.685 - 1] = [0.315, -0.315] \quad \text{Node 4: } [0.746 - 1, 0.254 - 0] = [-0.254, 0.254]$$

$$\frac{\partial \mathcal{L}}{\partial W^{(1)}} = M^T \frac{\partial \mathcal{L}}{\partial Z} \approx \begin{bmatrix} 0.093 & -0.093 \\ -0.037 & 0.037 \end{bmatrix}, \quad W \leftarrow W - \eta \frac{\partial \mathcal{L}}{\partial W}$$

This completes one training iteration of the Vanilla GCN.